

Project Executable Files

Date	7 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

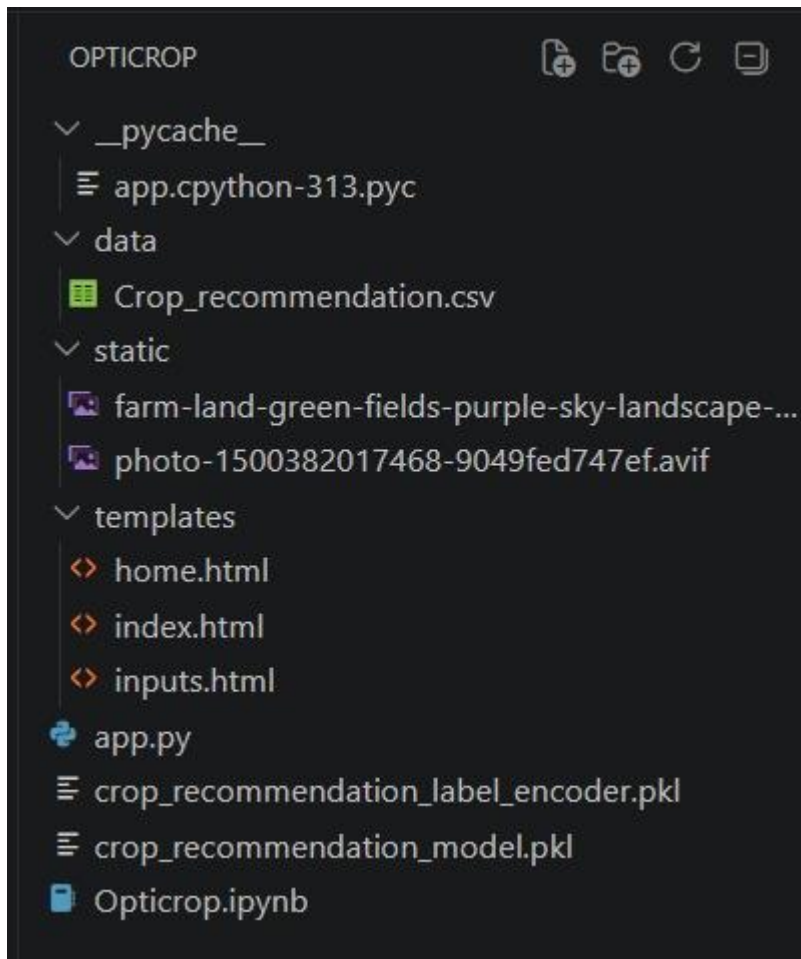
Project Executable Files

This document contains all the executable and deployable components of the **Smart Agricultural Production** project. The project is developed using Python, Flask, HTML, CSS, JavaScript, and Machine Learning. All source files, datasets, trained model files, templates, static resources, and dependency files are organized in a structured directory, allowing evaluators to run the application with minimal setup.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable)	NA
8	Dockerfile / Containerization configuration	NA
9	Test files and test results	Yes
10	Demo video or walkthrough	Yes

Step 2: File / Folder Structure



Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted (Runs Locally)
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Development Server
Repository Link	Git-hub
Demo Video Link	Demo-link

Step 4: Run Instructions

Pre-requisites

- Python 3.10 or later
- Flask
- NumPy
- Pandas
- Scikit-learn
- Pickle (Python Standard Library)

Steps

Step 1: Download or clone the project source code.

Step 2: Open the project folder in Visual Studio Code or any Python IDE.

Step 3: Install the required dependencies using:

```
pip install -r requirements.txt
```

Step 4: Ensure that the trained machine learning model (`model.pkl`) and dataset are present in the project directory.

Step 5: Run the Flask application:

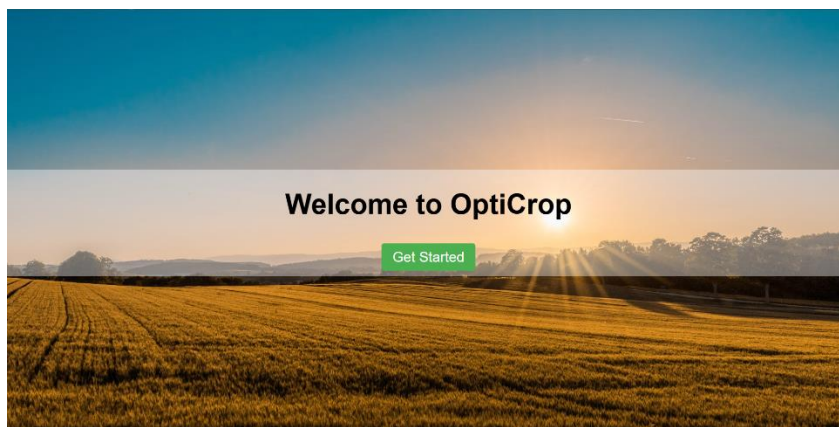
```
python app.py
```

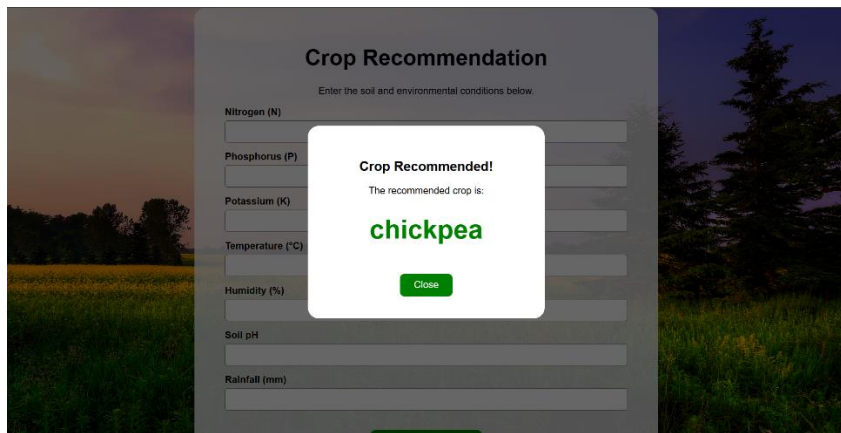
Step 6: Open a web browser and navigate to:

```
http://127.0.0.1:5000/
```

Step 7: Enter the required soil and environmental parameters, then click **Predict** to view the recommended crop.

```
PS C:\Users\sande\OneDrive\Desktop\GuessWhat\Personal\projects\Opticrop> C:\Users\sande\OneDrive\Desktop\django\environments\opticrop\Scripts\activate.ps1
(Opticrop) PS C:\Users\sande\OneDrive\Desktop\GuessWhat\Personal\projects\Opticrop> flask --app app run
C:\Users\sande\OneDrive\Desktop\django\environments\opticrop\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator DecisionTreeClassifi
er from version 1.6.1 when using version 1.9.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
C:\Users\sande\OneDrive\Desktop\django\environments\opticrop\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator RandomForestClassifi
er from version 1.6.1 when using version 1.9.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
C:\Users\sande\OneDrive\Desktop\django\environments\opticrop\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator LabelEncoder from ve
rsion 1.6.1 when using version 1.9.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
```





Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	The application requires manually entered input values.	Future enhancement will integrate IoT sensors for automatic data collection.
2	Real-time weather information is not integrated.	Future enhancement will include Weather API integration.
3	The application currently runs on a local server only.	Can be deployed on cloud platforms such as Render, Railway, or PythonAnywhere in future versions.